# A9126 Flash Strap

*User Guide*

**DMP Electronics Inc.**

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# Revision History

| REV. | Date | History |
|---|---|---|
| REV 01 | 2014/10/30 | 1. Initial Version. |
| REV 02 | 2015/03/18 | 1. Add section 4 for two examples. |
| REV 03 | 2015/05/25 | 1 Add section 2.6 for Dual or Signal CPU setting.<br>2 Modify and add more types on table 3. |
| REV 04 | 2017/07/14 | 1. Modify the figure 2 on section 2.2 for DRAM Address Mirror. |



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# 1. Overview

This is A9126 Flash strap user guide. This document will help you to setting the A9126 CPU、DRAM and PCI frequency at flash strap.

# 2. Flash strap setting

## 2.1. Strap type

By Flash (CPU/DRAM/PCIBOARD ID)

Flash Strap Location: FFFB6h – FFFB7h, FFFBBh – FFFBFh

Flash Format (7 Bytes)

| 7 | 6 | 5 | 4 | 3 | 2 | 1 | 0 | |
|---|---|---|---|---|---|---|---|---|
| CPU_NS | | | | | | | | Byte0(FFFB6h) |
| DRAM_NS | | | | | | | | Byte1(FFFB7h) |
| DRAM_MS | | | CPU_RS | | CPU_MS | | | Byte2(FFFBBh) |
| SPI_DIV[1] | PCI_DIV | | | | | DRAM_RS | | Byte3(FFFBCh) |
| SPI_DIV[3:2] | | CPU1 Power Down | SPI_DIV[0] | DRAM Addr-Cmd 2T | DRAM Addr mirror | RSVD | | Byte4(FFFBDh) |
| DRAM 2RANK | BOARD ID(L) | | | | | | | Byte5(FFFBEh) |
| Version ID | | | BOARD ID(H) | | | | | Byte6(FFFBFh) |



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## 2.2. What is “DRAM Addr mirror” at flash byte4[2]?

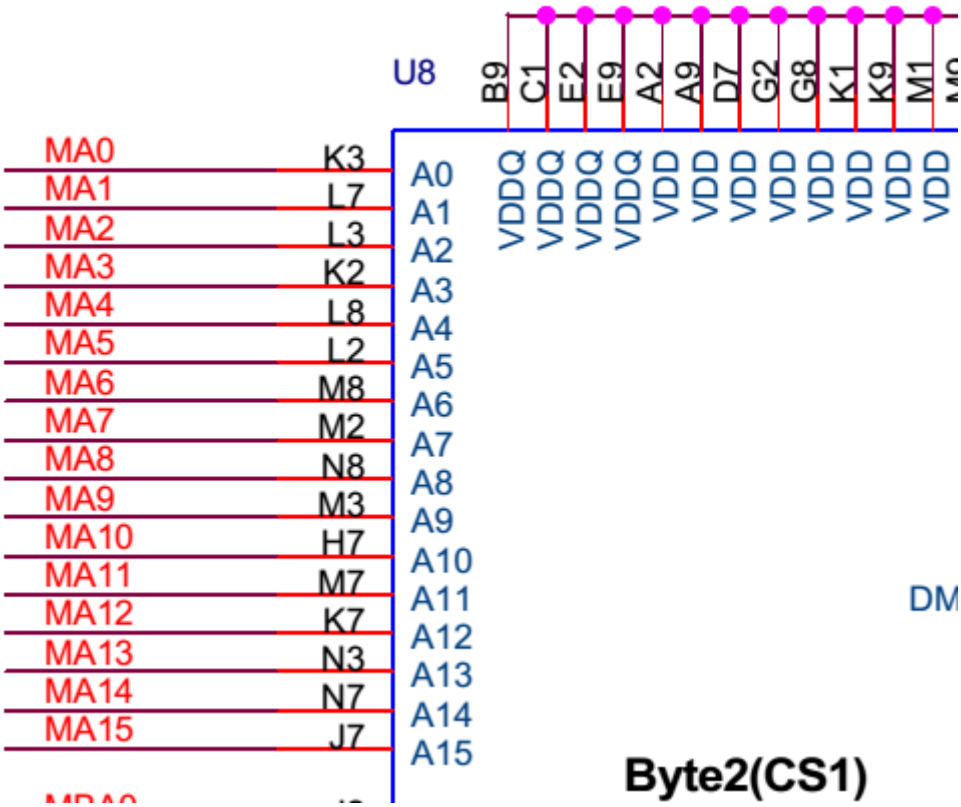
If your PCB schematic is design with DRAM Address mirror as Figure 2, please set “DRAM Addr mirror at flash byte4[2]”.

DRAM Address Mirror(depend on flash byte2[7:5])

0: Disable (default, if NB F0 60h bit[30]=0)

1: Enable

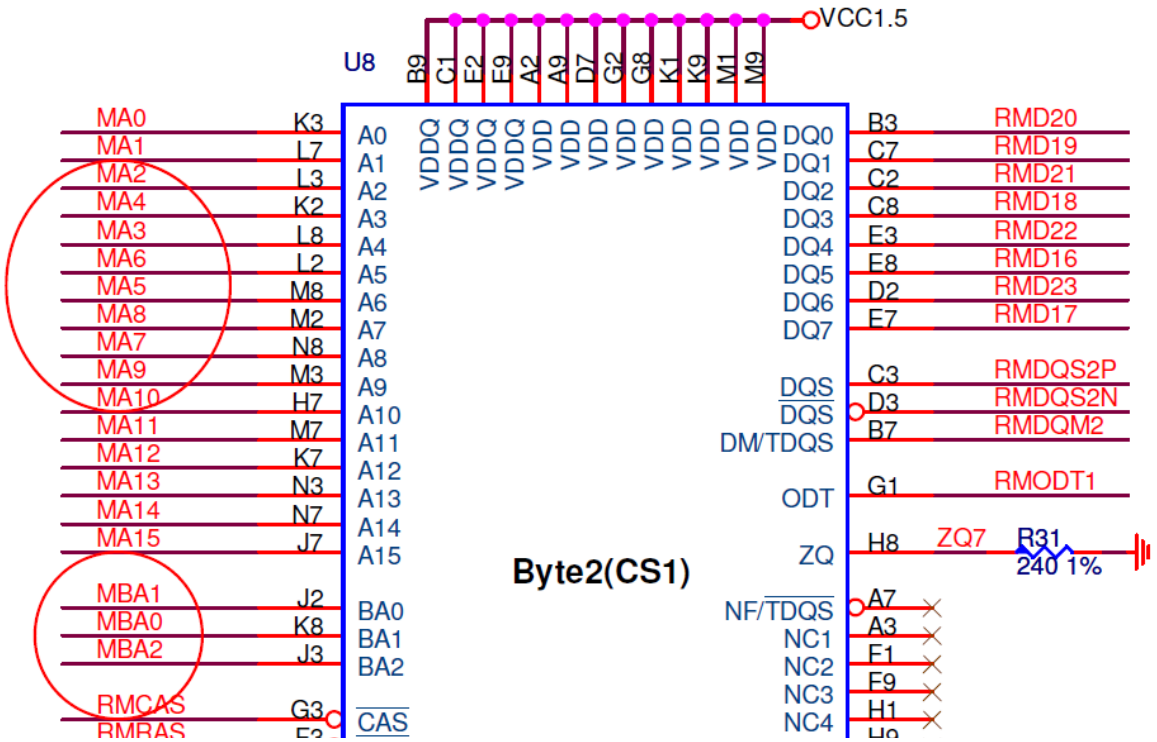
**Figure 1. DRAM without Address Mirror**






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**Figure 2 DRAM with Address Mirror**






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## 2.3. What is “DRAM Addr-Cmd 2T” at flash byte4[3]?

When the DRAM is unstable, you may try to set this bit.

But, the DRAM address command will increased 1T clock.

DRAM Addr-Cmd 2T (depend on flash byte2[7:5])

0: Disabled (default, if North Bridge Function 0 60h [30]=0)

1: Enabled

## 2.4. What is “DRAM 2RANK” at flash byte5[7]?

DRAM 2 Rank Selection (depending on flash byte5[7] )

0: Single Rank

1: Dual Rank (default)



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## 2.5. CPU/SPI_DIV/PCI_DIV table

**Table 1. CPU Frequency、SPI_DIV and PCI_DIV**

| CPU/PCI Frequency | CPU_MS BBh[0:2] | CPU_RS BBh[3:4] | CPU_NS B6h[0:7] | SPI_DIV[0] BDh[4] | SPI_DIV[1] BCh[7] | SPI_DIV[2:3] BDh[6:7] | PCI_DIV BCh[2:6] |
|---|---|---|---|---|---|---|---|
| 600/33MHz | 3'b011 | 2'b01 | 90h | 1'b0 | 1'b1 | 2'b01 | 5'b00111 |
| 700/33MHz | 3'b011 | 2'b01 | A8h | 1'b1 | 1'b1 | 2'b01 | 5'b01000 |
| 800/33MHz | 3'b011 | 2'b01 | C0h | 1'b0 | 1'b0 | 2'b10 | 5'b01010 |
| 800/66MHz | 3'b011 | 2'b01 | C0h | 1'b0 | 1'b0 | 2'b10 | 5'b00100 |
| 800/100MHz | 3'b011 | 2'b01 | C0h | 1'b0 | 1'b0 | 2'b10 | 5'b00010 |
| 900/33MHz | 3'b011 | 2'b00 | 6Ch | 1'b1 | 1'b0 | 2'b01 | 5'b01011 |
| 900/66MHz | 3'b011 | 2'b00 | 6Ch | 1'b1 | 1'b0 | 2'b01 | 5'b00101 |
| 900/100MHz | 3'b011 | 2'b00 | 6Ch | 1'b1 | 1'b0 | 2'b01 | 5'b00011 |
| 1.0/33 GHz | 3'b011 | 2'b00 | 78h | 1'b1 | 1'b0 | 2'b01 | 5'b01101 |
| 1.0/66 GHz | 3'b011 | 2'b00 | 78h | 1'b1 | 1'b0 | 2'b01 | 5'b00101 |
| 1.0/100 GHz/MHz | 3'b011 | 2'b00 | 78h | 1'b1 | 1'b0 | 2'b01 | 5'b00011 |
| 1.1/ 33 GHz/MHz | 3'b011 | 2'b00 | 84h | 1'b0 | 1'b1 | 2'b01 | 5'b01110 |
| 1.1/100 GHz/MHz | 3'b011 | 2'b00 | 84h | 1'b0 | 1'b1 | 2'b01 | 5'b00011 |
| 800/33MHz (HarewareDefault) | 3'b011 | 2'b00 | 90h | 1'b0 | 1'b1 | 2'b01 | 5'b10000 |



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**Table 2. DRAM Frequency**

| DRAM Frequency | DRAM_MS B6h[5:6] | DRAM _RS B6h[0:1] | DRAM _NS B7h[0:7] |
|---|---|---|---|
| 400 MHz (Hareware Default) | 3'b011 | 2'b01 | 60h |
| 533 MHz | 3'b011 | 2'b01 | 80h |
| 667 MHz | 3'b011 | 2'b01 | A0h |
| 800 MHz | 3'b011 | 2'b01 | C0h |
| 933 MHz | 3'b011 | 2'b00 | 70h |
| 1066MHz | 3'b011 | 2'b00 | 80h |

## 2.6. What is "CPU1 Power Down" at flash byte4[5]?

CPU1 Power Down Selection for signal core of CPU

0: CPU1 Power-Up (Dual Core of CPU)

1: CPU1 Power-Down (Signal Core of CPU)



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## 3. CPU/DRAM/PCI frequency table for DX3

**Table 3. CPU、DRAM and PCI frequency table**

| | Flash type | | | | | | |
|---|---|---|---|---|---|---|---|
| **CPU/DRAM/PCI** | **B6h** | **B7h** | **BBh** | **BCh** | **BDh** | **BEh** | **BFh** |
| 1G/800/33<br>NO Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | C0h | 63h | 35h | 50h | 80h | **1Dh** |
| 1G/800/33<br>Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | C0h | 63h | 35h | 54h | 80h | **1Dh** |
| 1G/800/33<br>NO Addr mirror<br>No Add-Cmd 2T<br>1Rank | 78h | C0h | 63h | 35h | 50h | 00h | **1Dh** |
| 1G/800/33<br>**Signal Core**<br>Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | C0h | 63h | 35h | 74h | 80h | **1Dh** |
| 1G/667/33<br>No Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | A0h | 63h | 35h | 50h | 80h | **1Dh** |
| 1G/667/33<br>Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | A0h | 63h | 35h | 54h | 80h | **1Dh** |



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| | Flash type | | | | | | |
|---|---|---|---|---|---|---|---|
| **CPU/DRAM/PCI** | **B6h** | **B7h** | **BBh** | **BCh** | **BDh** | **BEh** | **BFh** |
| 900/533/33<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | 6Ch | 80h | 63h | 2Dh | 54h | 80h | **1Dh** |
| 900/533/**100**<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | 6Ch | 80h | 63h | **0Dh** | 54h | 80h | **1Dh** |
| **800**/533/33<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | **C0h** | 80h | **6Bh** | **29h** | **44h** | 80h | **1Dh** |
| **800**/533/**100**<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | **C0h** | 80h | **6Bh** | **09h** | **44h** | 80h | **1Dh** |
| 800/**400**/33<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | C0h | **60h** | 6Bh | **29h** | **44h** | 80h | **1Dh** |
| 800/**400**/**100**<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | C0h | **60h** | 6Bh | **09h** | **44h** | 80h | **1Dh** |
| **600**/400/33<br>Address Mirror<br>No Add-Cmd 2T<br>2 Rank | **90h** | 60h | 6Bh | **9Dh** | **14h** | 80h | **1Dh** |



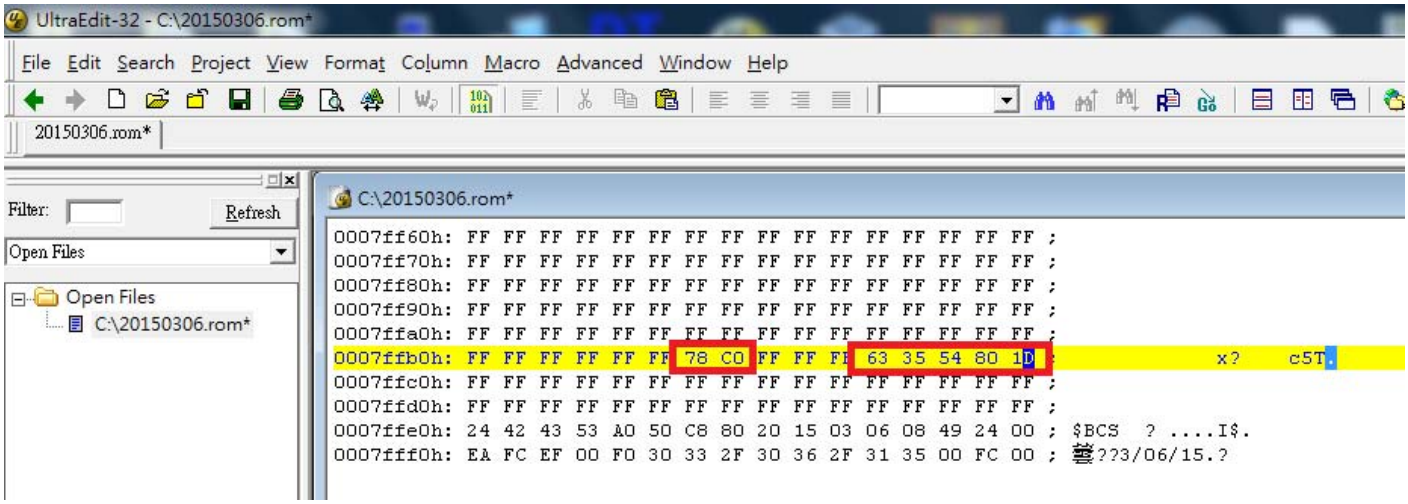
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## 4. How to set the flash strap and install BIOS file onboard

***Example 1:***

| | Flash type | | | | | | |
|---|---|---|---|---|---|---|---|
| **CPU/DRAM/PCI** | **B6h** | **B7h** | **BBh** | **BCh** | **BDh** | **BEh** | **BFh** |
| 1G/800/33<br>Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | C0h | 63h | 35h | 54h | 80h | **1Dh** |

(1) Open BIOS ROM by UltraEditor-32 or similar prgram, and then go to 0007ffb6h~0007ffbfh and set the value as top figure. Last, save the BIOS as other file name or save directly.




(2) Copy BIOS file upon a DOS bootable of USB mass storage.

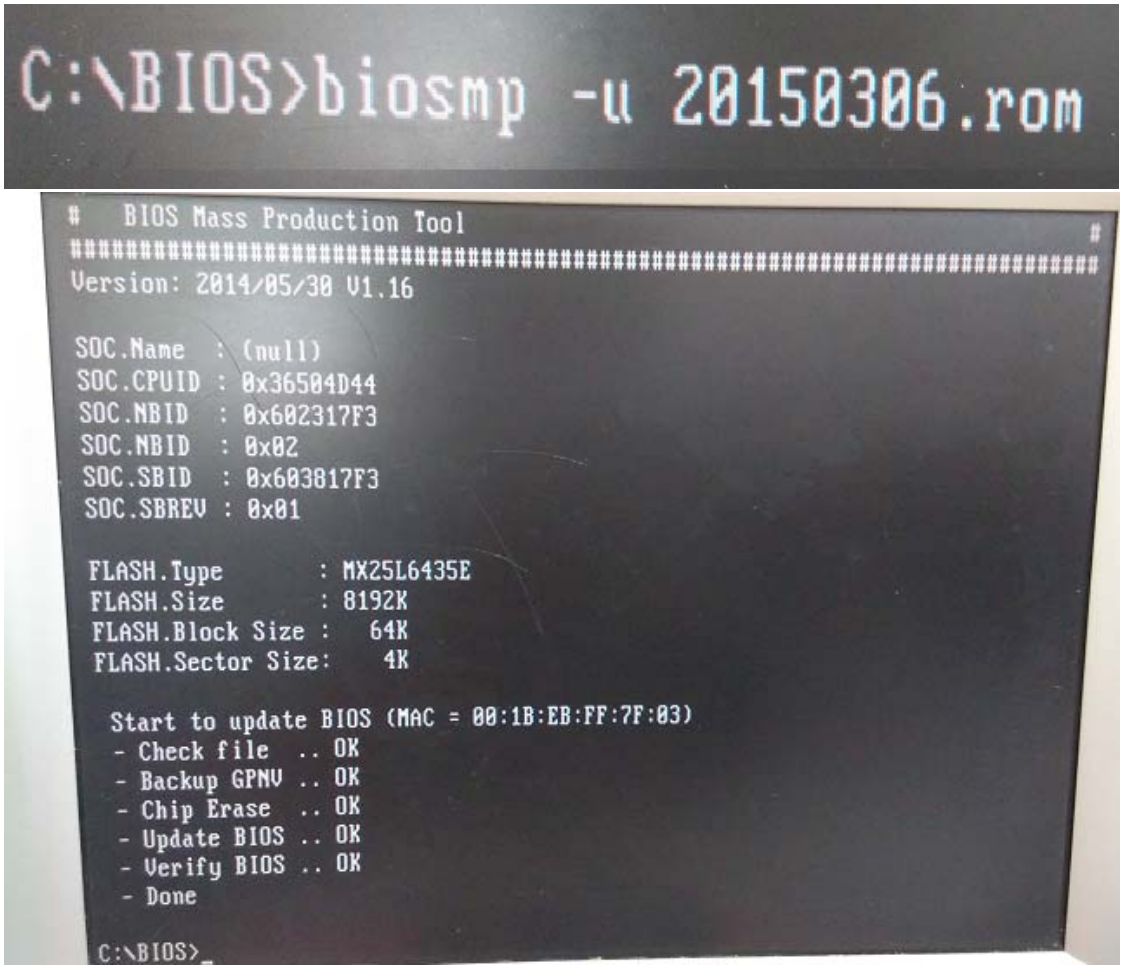
(3) Plug that USB mass stroage on VDX3 board.

(4) Power on VDX3 board and then press "F10" to choose first boot device to USB mass storage.



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(5) In DOS, please execute "BIOSMP -U XXX.ROM" for BIOS updating as below.



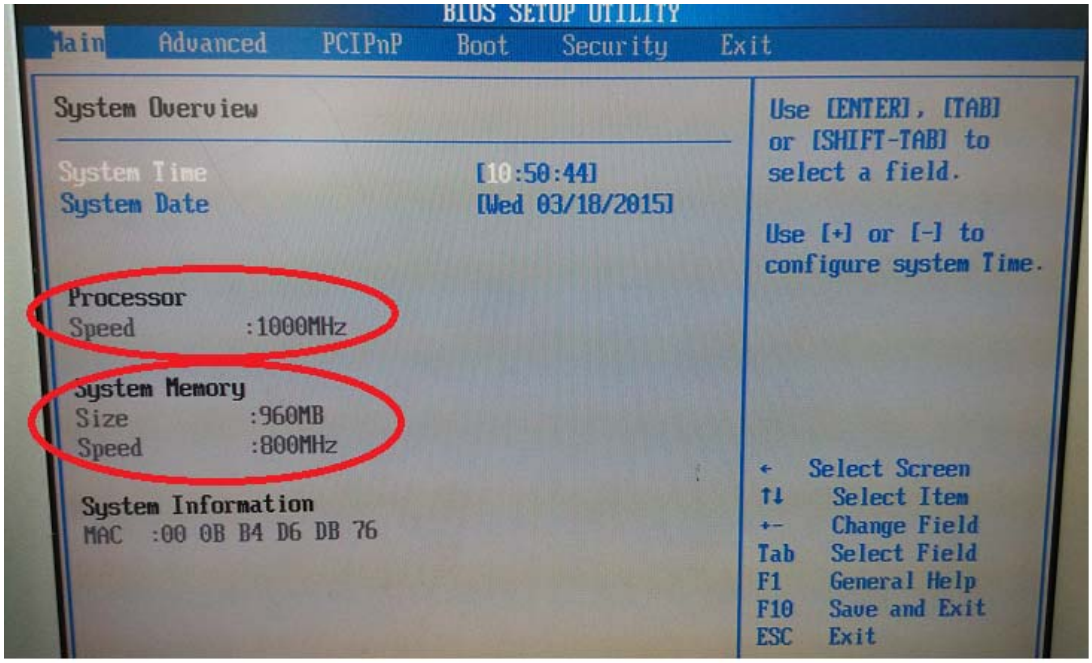

(6) After finish, please reboot and then press "Del" key into the BIOS setup.

(7) In BIOS setup, please press "F8" key to load failsafe defaults and then press "F9" to load optimal defaults.



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(8) You can see CPU/DRAM frequency is 1GHz/800MHz.



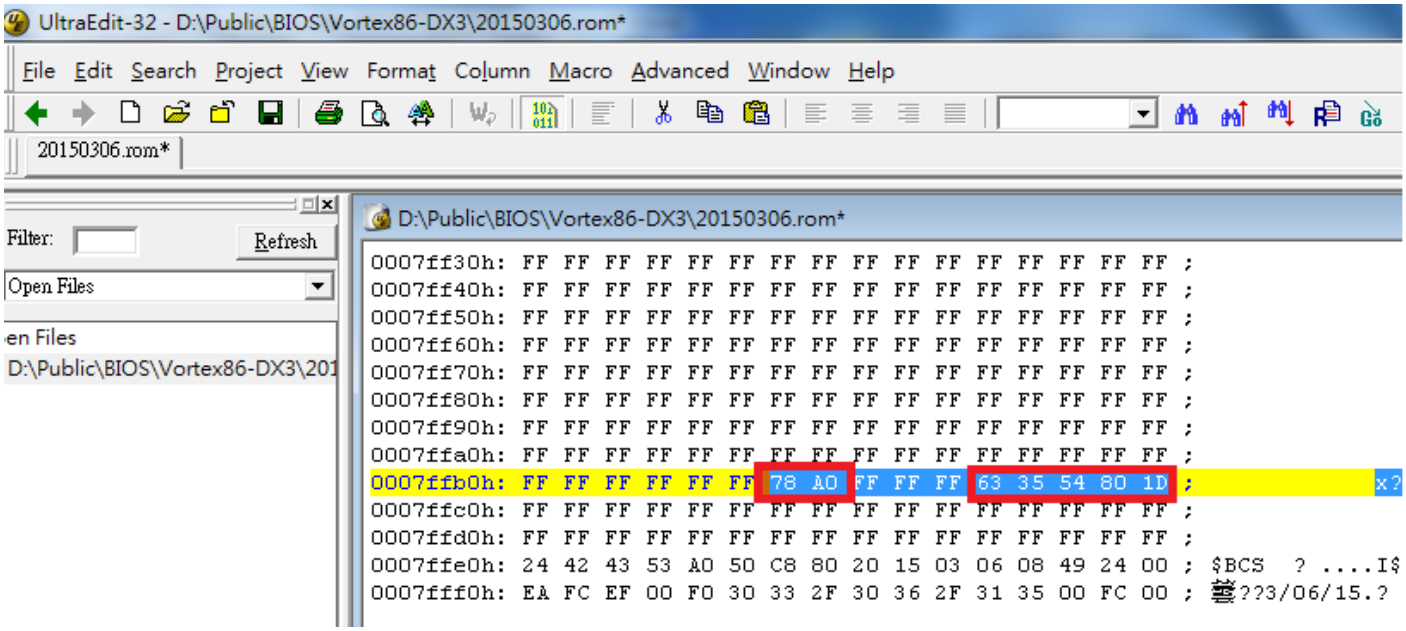

(9) Press "F10" to save the changes and exit.


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***Example 2:***

| | Flash type | | | | | | |
|---|---|---|---|---|---|---|---|
| **CPU/DRAM/PCI** | **B6h** | **B7h** | **BBh** | **BCh** | **BDh** | **BEh** | **BFh** |
| 1G/667/33<br>Addr mirror<br>No Add-Cmd 2T<br>2Rank | 78h | **A0h** | 63h | 35h | 54h | 80h | **1Dh** |

(1) Open BIOS ROM by UltraEditor-32 or similar prgram, and then go to 0007ffb6h~0007ffbfh and set the value as top figure. Last, save the BIOS as other file name or save directly.




(2) Copy BIOS file upon a DOS bootable of USB mass storage.

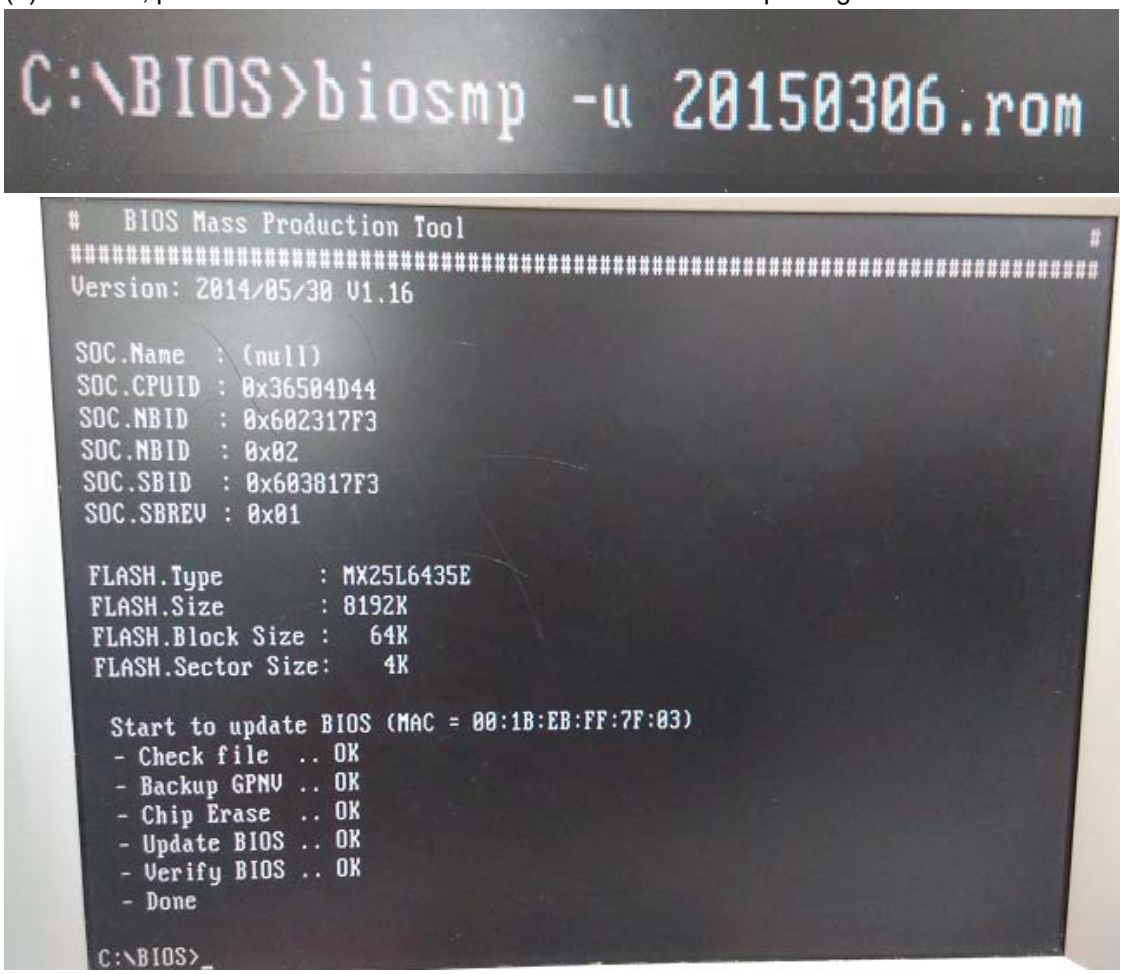
(3) Plug that USB mass stroage on VDX3 board.

(4) Power on VDX3 board and then press "F10" to choose first boot device to USB mass storage.



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(5) In DOS, please execute "BIOSMP -U XXX.ROM" for BIOS updating as below.



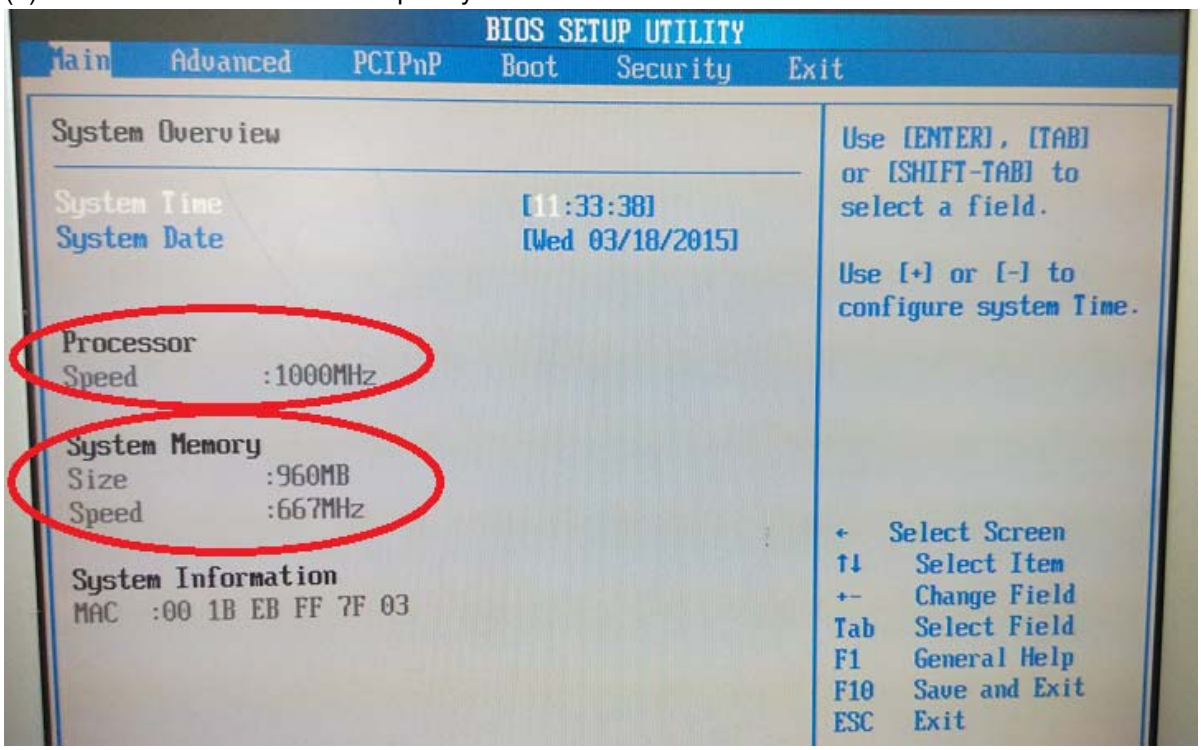

(6) After finish, please reboot and then press "Del" key into the BIOS setup.

(7) In BIOS setup, please press "F8" key to load failsafe defaults and then press "F9" to load optimal defaults.



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(8) You can see CPU/DRAM frequency is 1GHz/667MHz..




(9) Press "F10" to save the changes and exit.


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